

Mathematician with a strong command of probability, stochastic modelling and mathematical finance. Experience in statistical modelling and operationalizing machine learning workflows.

PhD – Mathematical Finance
Central European University 2018–2022

MSc – Applied Mathematics
Budapest University of
Technology and Economics 2015–2018

BSc – Applied Mathematics
SZTE, Bolyai Institute 2012–2015



Professional Experience

Eötvös Loránd University, ELTE AI Research Group

2023–Present*Research Assistant*

- Research deep-learning methods for statistical inference in stochastic-process and long-memory time-series models.
- Conducted an interpretability study on neural networks applied to time-series data: investigated which characteristics of the input drive the networks' inference.
- Led the technical development of reinforcement-learning methods for automated trading in stylized market environments, driving the project from problem formulation through model development to experimental validation.

Alfréd Rényi Institute of Mathematics

2019–Present*Research Fellow*

- Investigate alternative stochastic dynamics for diffusion-based generative models.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Artificial Intelligence National Laboratory

2024*ML Researcher*

- Collaborated with ALTEO on intraday electricity-price forecasting.
- Developed and evaluated machine-learning regression models, with a focus on predictive reliability.

Sysler Kft.

2023*Research Consultant*

- Prototyped a Bayesian late-fusion framework combining outputs from independent medical-diagnostic models to infer disease progression from empirical patient trajectories.

Morgan Stanley

2017*Interest Rate Modelling Intern*

- Reviewed financial-product documentation, checking model assumptions, calculations, pricing results, and internal consistency.
- Reverse-engineered parts of a legacy pricing system written in A+ to understand existing model and pricing logic.

Technical Skills

- Expertise in statistical **analysis**, statistical learning, **probability** theory, stochastic **optimization**, mathematical finance, information theory
- Experience in machine learning operations, version control with Git, working knowledge of CI/CD, containerizing applications, automated training, evaluation, and experiment tracking, hyperparameter optimization (Docker, Azure Machine Learning, TensorBoard, Optuna, MLflow, PyTorch, GitHub Actions)
- Strong understanding of the *mathematical* foundations of classical and modern **machine learning methods**: reinforcement learning, ensemble models, late fusion, Bayesian filtering, variational inference, diffusion generative models